DOMPERIDONE

QT prolongation

Max 7 days

CHEMICAL NAME 5-chloro-1-{1-[3-(2-oxo-2,3-dihydro-1H-benzimidazol-1-yl)propyl]piperidin-4-yl}-1,3-dihydro-2H-benzimidazol-2-one

CLASS Peripheral dopamine D2 antagonist antiemetic and prokinetic

TRADE NAMES Motilium · Motinorm · Domstal · Costi

THERAPEUTIC USES / INDICATIONS

- Relief of nausea and vomiting (short-term).
- Nausea with dopamine agonists in Parkinson's disease (does not worsen parkinsonism).
- Gastroparesis (off-label); lactation stimulation (off-label).

ADMINISTRATION

- **PO** 15–30 min before meals.
- Use lowest dose for max 7 days.

DOSAGES

- **Adult & ≥35 kg:** 10 mg up to three times daily; max 30 mg/day.
- **Children <35 kg:** 0.25 mg/kg up to three times daily.
- Avoid in moderate–severe hepatic impairment; reduce frequency in severe renal impairment.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** 4–6 h.
- Low bioavailability (~15%); metabolised by CYP3A4.
- Excreted in faeces. **t½:** 7–9 h.

PHARMACODYNAMICS

- Blocks **D2 receptors** in CTZ (outside BBB) and gut → antiemetic and prokinetic.
- Does not cross blood–brain barrier → few extrapyramidal effects.
- ↑ prolactin.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth, headache.
- Abdominal cramps, diarrhoea.
- Galactorrhoea, gynaecomastia.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, ventricular arrhythmias, sudden cardiac death (higher risk >60 yr, >30 mg/day).
- Extrapyramidal effects (rare, infants).

CONTRAINDICATIONS & CAUTIONS

- Cardiac disease, QT prolongation, significant electrolyte disturbance.
- Moderate–severe hepatic impairment.
- Use with strong CYP3A4 inhibitors or QT-prolonging drugs.
- GI obstruction, perforation, haemorrhage; prolactinoma.

KEY INTERACTIONS

- **Ketoconazole, erythromycin, clarithromycin, ritonavir:** contraindicated.
- **QT drugs:** avoid.
- **Anticholinergics:** antagonise prokinetic effect.

NURSING MANAGEMENT

- Check cardiac history, ECG risk, electrolytes.
- Review interacting medicines.
- Limit duration; give before meals.

PATIENT EDUCATION

- Take before meals for up to a week.
- Report palpitations, fainting.
- Do not exceed 30 mg/day.

KEY POINTS

- Peripheral D2 antagonist; no EPS.
- QT prolongation — avoid in heart disease.
- Max 10 mg TDS for 7 days.
- Useful in Parkinson's nausea.

SOURCES CHECKED **[533]** Domperidone 1mg/ml Oral Suspension (UK SmPC)

Full references p. 432–457